

Seminar on Nature-Based Solutions
CUNY EES 79903 (3 credits)
Fall 2026 Syllabus

Course Catalog Description: Discussion of nature-based solutions in the context of urban water resource management; investigation of hybrid systems of built and natural features in cities; exposure to careers in urban environmental management.

Logistics:

Time: Tuesdays 4:15-6:15pm
Location: CUNY Grad Center (365 5th Ave), Room 3309
Instructors: Kyle McKay, Ph.D., P.E. Brett Branco, Ph.D.
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970-980-9747 917-612-4293
Office – CUNY Grad Center, Room 4439
Office hours on Mondays, Tuesdays, or Thursdays by appointment
Pre-requisites: None
Course website: <https://skmckay.github.io/CUNY.NBS.2026/>

Course Objectives

This seminar-based course will explore the growing use of nature-based solutions in cities, which are often proposed in the context of simultaneously managing multiple crises related to biodiversity loss, climate change, aging infrastructure, and inequity of natural hazards. Students will explore these topics through reading peer-reviewed literature, agency and project reports, and other resources on the subject. The instructors will guide the class through a critical examination of NBS and their efficacy for accomplishing societal goals, challenges in project execution, and scientific uncertainties. When possible, discussions will include practitioners working in this space, who can provide insights on career trajectories, personal experiences, and disciplinary expertise. This interdisciplinary-oriented course is intended for students from multiple disciplines including environmental sciences, physical sciences, ecology, environmental design, natural resources, marine sciences, anthropology, economics, engineering, and other relevant and related fields. Upon completion of this course, students will be able to:

- Define and describe nature-based solutions for addressing urban sustainability challenges.
- Understand the role of both ecological and built systems in cities.
- Critically analyze the complex interactions between natural systems and built infrastructure within the context of water resource management.
- Foster an interdisciplinary approach (and dialog) by integrating knowledge from diverse fields to enhance natural infrastructure practices.

Responsibilities and Expectations

The faculty and administration of Brooklyn College support an environment free from cheating and plagiarism. Each student is responsible for being aware of what constitutes cheating and plagiarism and for avoiding both. If a faculty member suspects a violation of academic integrity and, upon investigation, confirms that violation, or if the student admits the violation, the faculty member MUST report the

violation. Students should be aware that faculty may use plagiarism detection software. The complete text of the CUNY Academic Integrity Policy can be found at:

<https://www.cuny.edu/about/administration/offices/legal-affairs/policies-resources/academic-integrity-policy/>

Class attendance is expected, unless arranged otherwise with prior notice. Missed classes will count against your grade without notification. We also expect the following general behaviors to facilitate productive time together.

- Treat all class participants respectfully to create a safe, equitable, inclusive, supportive, and (ultimately) more productive environment that is free of harassment and discrimination, in which all feel welcome and valued.
- Read all assigned papers and participate actively in discussions. Lead discussion on a minimum of 3 papers over the course of the semester. Typically discussions will be led in conjunction with other student leaders for the week.

Grading System

Grades will be allocated for participation and attendance (50%), effective discussion leadership (25%), and the class project (25%). Grades are defined as: A > 90%, B = 80-89.99%, C = 70-79.99%, D = 60-69.99%, and F < 60%. All grades are final, and assignments may not be revised or resubmitted.

Weekly Course Format

- Before class each week, students will read 2-3 assigned papers or reports on a focal subject related to nature-based solutions.
- The instructors will present a brief lecture to contextualize reading materials along with a current event related to NBS. These may include case studies, ongoing research briefs, visitors from other institutions, and/or important happenings in the field.
- Students will lead and participate in a discussion of the assigned reading.

Class Project

During the semester, students will contribute to a “state of the art” summary document focused on coastal natural and nature-based features. The document will be modeled after the format of climate assessments, which often include a set of key messages on topics with a “traceable account” supporting the key messages. In this context, a traceable account provides the rationale for the key message, including the literature supporting it. Here, we can use a format fashioned after the NYS Climate Assessment:

<https://nysclimateimpacts.org/about-the-assessment/methods-and-sources/>

Key findings represent important takeaways identified by the student teams. Key findings generally focus not only on problems but also impacts, responses, and solutions. They attempt to convey:

- The “What?” Current challenges, circumstances, facts, observations, and projections.
- The “So what?” The implications and meaning, including underlying assumptions.
- The “Now what?” Potential actions, trade-offs, and resources needed to address the challenges, as well as knowledge gaps.

Traceable accounts provide supporting evidence for each of the key findings. They describe supporting information and the level of confidence in each finding, allowing interested readers to “trace” the rationale and information sources used to develop each finding. Traceable accounts must include:

- The evidence that supports each key finding, including references and sources
- The quality of the evidence
- The level of agreement in the literature
- Key uncertainties

Potential topics for the “state of the art” document for nature-based solutions are below. Each topic would include 3-5 key messages, with 1-2 paragraphs in the traceable account for each key message.

- Qualitative foundations of project planning: systems context, primary and secondary design objectives, risk tolerance
- Facilitation, engagement, and teaming: interdisciplinary dialog, ways of knowing, ladder of participation, facilitation, co-production methods
- Ecological analysis of NBS: levels of effort, project outcomes
- Economic analysis of NBS: levels of effort, project outcomes
- Co-benefits analysis of NBS: levels of effort, project outcomes
- Engineering analysis of NBS: performance, resilience, cost
- Alternatives analysis: structured decision making, management measures, alternative development, decision support, trade-off analysis
- Performance-based design: iterative workflows, integrated modeling, hazard resilience and reliability, adaptability
- Constructability: material selection, regulatory requirements, developing specifications, construction oversight
- Performance monitoring: outcomes, performance, cost
- Project establishment: adaptive management, transition to operations
- Asset management: condition tracking, operations, maintenance, repair, rehabilitation, decommissioning, life cycle budgeting

Background Resources (other materials will be provided prior to each discussion)

Bridges, T. S., J. K. King, J. D. Simm, M. W. Beck, G. Collins, Q. Lodder, and R. K. Mohan, eds. 2021. International Guidelines on Natural and Nature-Based Features for Flood Risk Management. Vicksburg, MS: U.S. Army Engineer Research and Development Center.

Downloadable at: <https://ewn.erd.cdrn.mil/international-guidelines-on-natural-and-nature-based-features-for-flood-risk-management/>

McPhearson T., Kabisch N., and Frantzeskaki N. 2023. Nature-based solutions for cities. Edward Elgar Publishing, Northampton, Massachusetts.

Downloadable at: <https://urbansystemslab.com/nbs-for-cities>

Outline of Topics

- Course intro + Brief history of nature-based solutions
- NBS fundamentals

- Structured decision making and community engagement
- Analysis of NBS: ecological outcomes, economic benefits, co-benefits, engineering performance
- Design of NBS: alternatives analysis, performance-based design, constructability, implementation
- Life cycle management: performance monitoring, natural asset management

Potential Reading List (subject to change)

- Abera L.E., Surbeck C.Q., and McKay S.K. 2024. Lifecycle cost and benefit analysis for parcel-scale implementation of green stormwater infrastructure. *Green Technologies and Sustainability* (100139) <https://doi.org/10.1016/j.grets.2024.100139>.
- Annis S., Badas M.G., and Mascaro G. 2026. Increasing the fidelity of hyperlocal simulations of urban pluvial flooding through street flooding observations. *Advance in Water Resources*.
- Anzelone M. and Silvera Seamans, G. (2025). *Oaks, Our City, and Us: A Vision for Nature in New York City*. NYC Biodiversity Task Force Report.
- Bredes, A., Tso, G., Gittman, R.K., Narayan, S., Tomiczek, T., Miller, J.K. and Morris, R.L., 2024. A 20-year systematic review of wave dissipation by soft and hybrid nature-based solutions (NbS). *Ecological Engineering*, 209, p.107418.
- Chambers M.L., Santiago-Collazo F.L., Lammers R.W., and Bledsoe B.P. 2026. A practitioner-friendly approach to quantifying the hydrologic benefits of city-scale distributed green stormwater infrastructure. *Journal of Environmental Management*, 409 (130012), <https://doi.org/10.1016/j.jenvman.2026.130012>.
- Chrysoulakis, N., Somarakis, G., Stagakis, S., Mitiraka, Z., Wong, M.S. and Ho, H.C., 2021. Monitoring and evaluating nature-based solutions implementation in urban areas by means of earth observation. *Remote Sensing*, 13(8), p.1503.
- Cohen-Shacham, E., Andrade, A., Dalton, J., Dudley, N., Jones, M., Kumar, C., Maginnis, S., Maynard, S., Nelson, C.R., Renaud, F.G. and Welling, R., 2019. Core principles for successfully implementing and upscaling Nature-based Solutions. *Environmental Science & Policy*, 98, pp.20-29.
- Cook, E.M., Kim, Y., Grimm, N.B., McPhearson, T., Anderson, P., Bulkeley, H., Collier, M.J., Diep, L., Morató, J. and Zhou, W., 2025. Nature-based solutions for urban sustainability. *Proceedings of the National Academy of Sciences*, 122(29), p.e2315909122.
- Crawford, B.A., Maerz, J.C. and Moore, C.T., 2026. Navigating multiple objectives, partners, and jurisdictions: Decision analysis for range-wide conservation planning for an at-risk species. *Conservation Science and Practice*, p.e70291.
- Curran J.C., McKay S.K., Beagle J.R., Brauer E.J., Haring C.P., Thomas J., Lange J., Duhr D., VanBuecken N., Rose C., Bishop D., and Jensen S. 2025. *Engineering practice guide for floodplain benching: A natural infrastructure approach for riverine systems*. ERDC TR-25-15. U.S. Army Engineer Research and Development Center, Vicksburg, Mississippi.
- Ding, H., and C. McLaren. 2025. *Financial sector guidebook on nature-based solutions investment: Aligning investment with impacts and showcasing examples*. Guidebook. Washington, DC: World Resources Institute. Available online at: <https://doi.org/10.46830/wriwb.24.00043>.
- Esquivel-Muelbert, J.R., Fontoura, L., Zawada, K., Erickson, K., Figueira, W., Madin, J.S. and Bishop, M.J., 2026. The natural architecture of oyster reefs maximizes recruit survival. *Nature*, pp.1-5.
- Feagin, R.A., Bridges, T.S., Bledsoe, B., Losos, E., Ferreira, S., Corwin, E., Lodder, Q., Beck, M.W., Reguero, B., Sutton-Grier, A. and Figlus, J., 2021. Infrastructure investment must incorporate Nature's lessons in a rapidly changing world. *One Earth*, 4(10), pp.1361-1364.
- González-García, A., Palomo, I., Codemo, A., Rodeghiero, M., Dubo, T., Vallet, A. and Lavorel, S., 2025. Co-benefits of nature-based solutions exceed the costs of implementation. *Cell Reports Sustainability*, 2(3).
- Government of the Hong Kong Special Administrative Region. 2026. *Hong Kong Nature-based Solutions Design Guidelines*. Civil Engineering and Development Department and Agriculture, Fisheries, and Conservation Department. Downloadable here.
- Hall, D.M. and van Rees, C.B., 2026. Implementing nature-based solutions requires distinguishing place from space. *Ambio*, pp.1-18.
- Heard S.G. 2026. How to lead a journal club you won't be embarrassed by later. *Scientist Sees Squirrel*. <https://scientistseesquirrel.wordpress.com/2026/01/13/how-to-lead-a-journal-club-you-wont-be-embarrassed-by-later/>.
- Ibrahim, A., Marshall, K., Carmen, E., Blackstock, K.L. and Waylen, K.A., 2025. Raising standards for stakeholder engagement in Nature-based Solutions: Navigating the why, when, who and how. *Environmental Science & Policy*, 163, p.103971.
- McKay, S.K., Wenger, S.J., van Rees, C.B., Bledsoe, B.P. and Bridges, T.S., 2023. Jointly advancing infrastructure and biodiversity conservation. *Nature Reviews Earth & Environment*, pp.1-3.
- McPhearson and Diep. 2025. *NATURA Global Roadmap for Urban Nature-based Solutions*. Website for download of the regional manuals.
- Natural Assets Initiative. 2024. *Natural is infrastructure: How to include natural assets in asset management plans. A guidebook for local governments*. Natural Assets Initiative.

Nesshöver, C., Assmuth, T., Irvine, K.N., Rusch, G.M., Waylen, K.A., Delbaere, B., Haase, D., Jones-Walters, L., Keune, H., Kovacs, E. and Krauze, K., 2017. The science, policy and practice of nature-based solutions: An interdisciplinary perspective. *Science of the total environment*, 579, pp.1215-1227.

Ostrow, K.S. and Cox, D.T., 2026. Performance-Based Design Methodology for Nature-Based Solutions: Using Mangroves to Mitigate Wave Overtopping. *ASCE OPEN: Multidisciplinary Journal of Civil Engineering*, 4(1), p.04026006.

Pathak A., Del Angel D.C., and Hale C. 2026. Co-developing nature-based solutions for coastal resilience: A case study from Refugio County, Texas. *Estuaries and Coasts*, 49, 31.

Pineda-Pinto, M., Kennedy, C., Nulty, F. and Collier, M., 2024. Leverage points for improving urban biodiversity conservation in the Anthropocene: A novel ecosystem lens for social-ecological transformation. *Environmental Science & Policy*, 162, p.103926.

Reidenbach, M.A., Li, M., Rose, K.A., Tomiczek, T., Morris, J., Palinkas, C.M., Staver, L.W., Nardin, W., Gray, M.W., Lee, S.B. and Sutton-Grier, A.E., 2025. Performance Evaluation of Natural and Nature-Based Features for Coastal Protection and Co-Benefits. *Annual Review of Marine Science*, 18.

Sadaf, A. and Bruck, J., 2026. Optimizing shoreline suitability models framework: integrating advanced GIS methodologies at Aberdeen Proving Ground, Maryland. *Frontiers in Environmental Science*, 14, p.1812979.

Silliman, B.R., Hensel, M.J., Temmink, R.J., Paxton, A.B., Wiczyński, D.J., Gibert, J.P., Johnston, D.W., Noonan, D., Ferris, W.N., Hay, M.E. and Paalman, T., 2026. Ecosystem technology (ecotech): Harnessing natural processes to address global challenges. *Science advances*, 12(19), p.eac5411.

Singhvi, A., Luijendijk, A.P. and van Oudenhoven, A.P., 2022. The grey-green spectrum: A review of coastal protection interventions. *Journal of Environmental Management*, 311, p.114824.

Spurgeon, S.L., Shawler, J.L., Smith, K.J., Marrero, L.I.B. and Jackson, S.S., 2026. Coastal nature-based solutions and remote sensing data: Selecting performance metrics and evaluating data availability. *Shore & Beach*, 94(1), p.28.

Stanford, G.O., Ferreira, S., Landry, C.E. and Blachly, B., 2025. Economic valuation of ecosystem services and natural infrastructure: A quantitative review of the literature. *Ecosystem Services*, 75, p.101754.

Suedel, B.C., Calabria, J., Bilskie, M.V., Byers, J.E., Broich, K., McKay, S.K., Tritinger, A.S., Woodson, C.B. and Dolatowski, E., 2022. Engineering coastal structures to centrally embrace biodiversity. *Journal of Environmental Management*, 323, p.116138.

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van Rees, C., Vanburen, C., Altieri, A., Bilskie, M., Byers, J., Calabria, J., Chambers, M., Coleman, D., Chambers, M.L., Foster, K. and Nelson, A.H., Leveraging Ecological Theory to Maximize Biodiversity of Nature-Based Solutions at Landscape Scales.

Webb B., Dix B., Douglass S., Asam S., Cherry C., and Buhring B. 2019. Nature-based solutions for coastal highway resilience: An implementation guide. FHWA-HEP-19-042. Federal Highway Administration, Washington, DC.

Wijsman, K., Auyeung, D.S., Brashear, P., Branco, B., Graziano, K., Groffman, P., Cheng, H. and Corbett, D., 2021. Operationalizing resilience: co-creating a framework to monitor hard, natural, and nature-based shoreline features in New York State. *Ecology and Society*, 26(3).

Wolff, E., Hamel, P., Lechner, A.M., Irvine, K., Frantzeskaki, N., Diep, L., Tavenisa, L., Welch, A., Cajilig, P.G., Sen, M. and Pottinger-Glass, C., 2026. Monitoring nature-based solutions is key for compliance, co-creation and practice improvement. *Nature Sustainability*, pp.1-10.